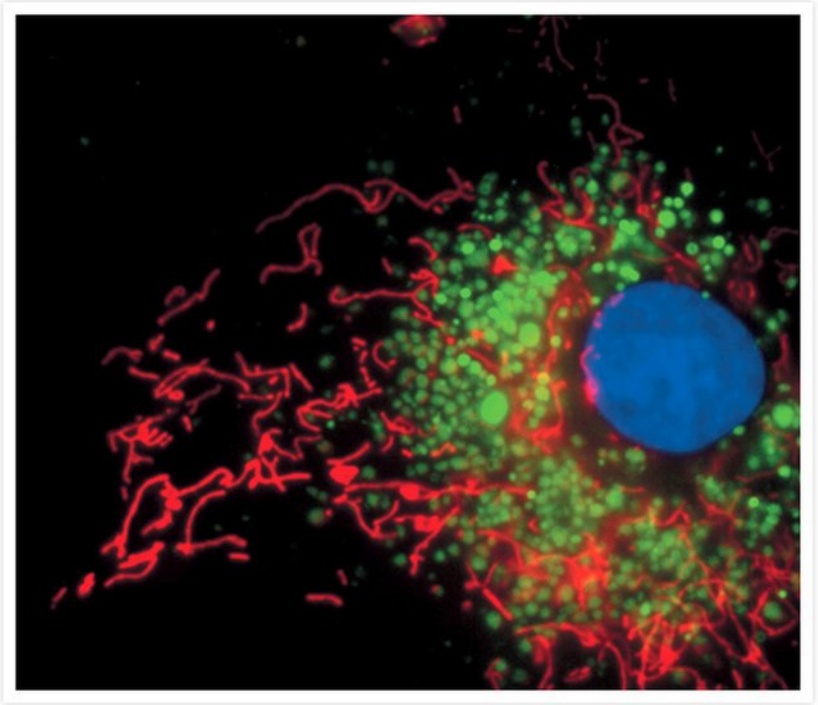
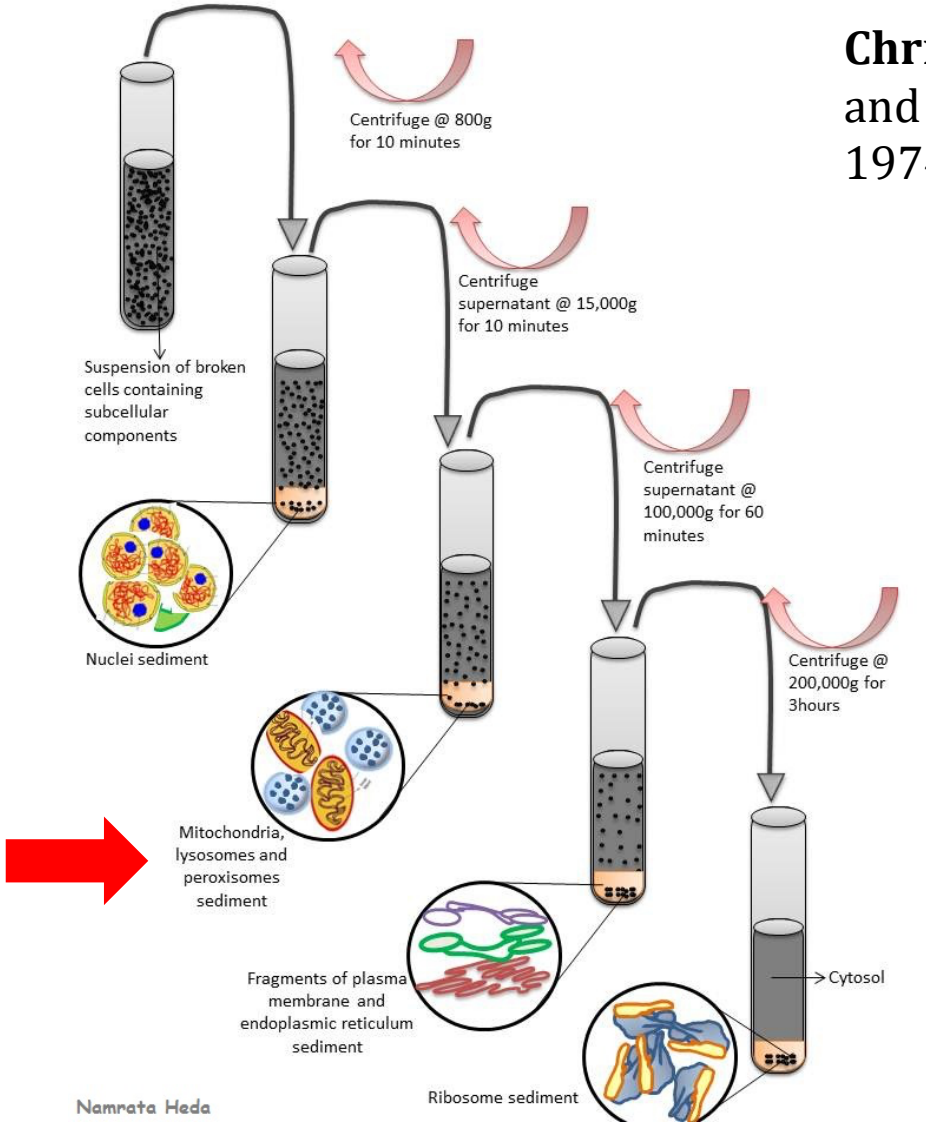
LYSOSOMES

OVERVIEW

- Discovery and structure
- Tagging and sorting lysosomal enzymes
- Lysosomal storage disorders



Discovery of lysosomes and peroxisomes: ultracentrifugation and TEM



Christian de Duve discovered lysosomes and peroxisomes. Nobel Prize in Medicine in 1974.

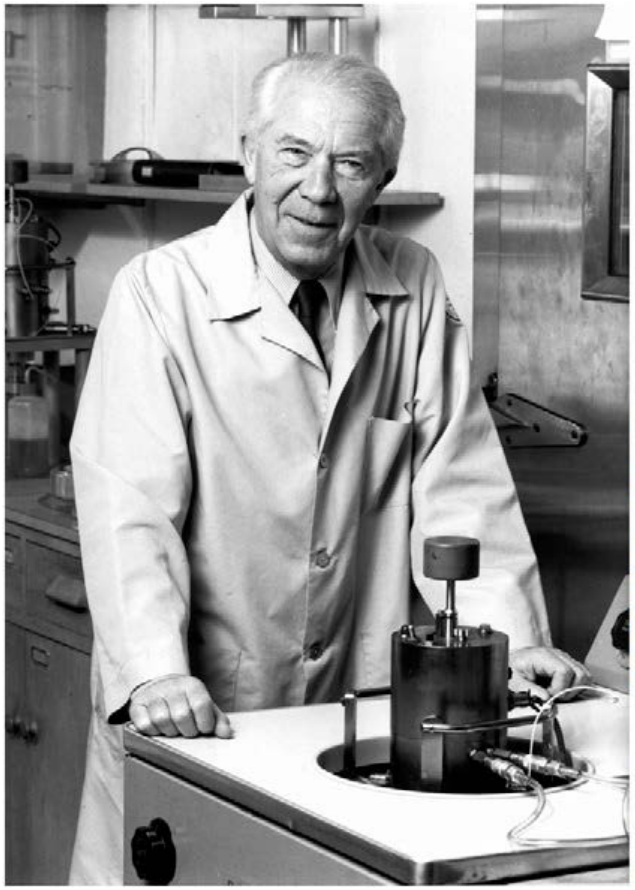
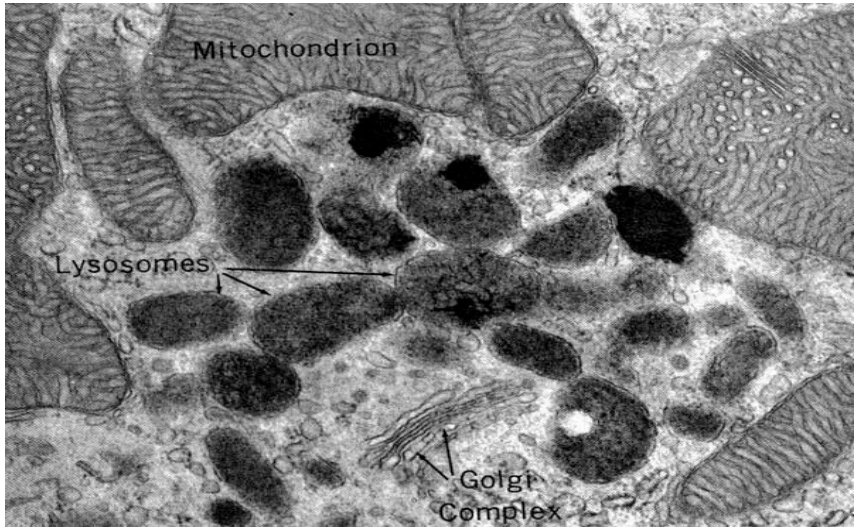


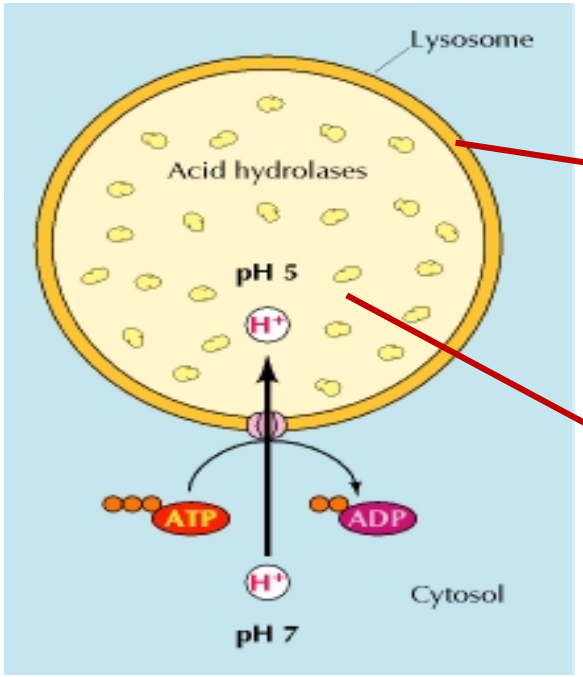
Photo courtesy of The Rockefeller University.

Structure of Lysosomes



Size: 0.2 to 0.5 μm

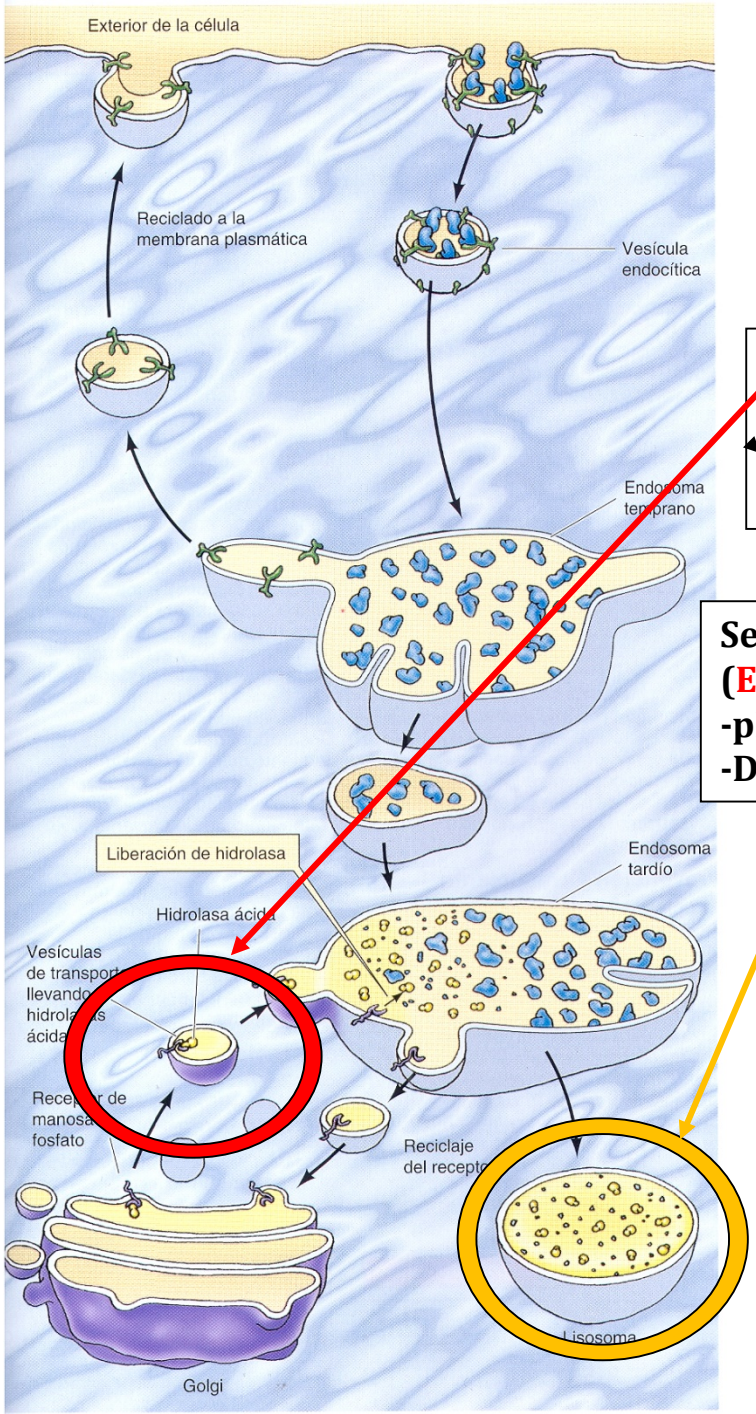
Lysosome: *lýs*(*is*) (decompose) + *sōma* (body)



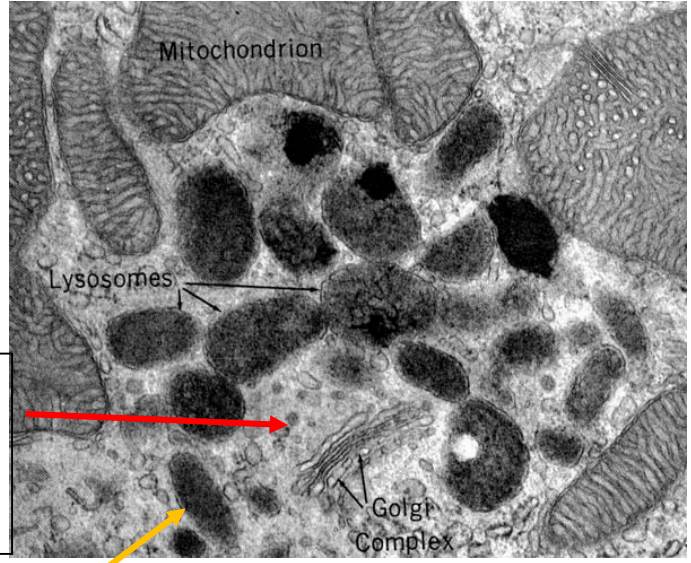
Membrane
- Lipid bilayer
- Carrier proteins
- Glycosylated proteins
- **Proton pumps**

Acid hydrolases: ~40 (lipases, nucleases…)

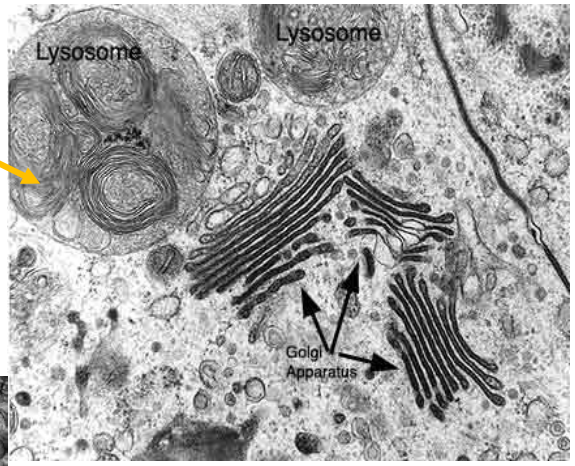
Acid phosphatase: a universal marker

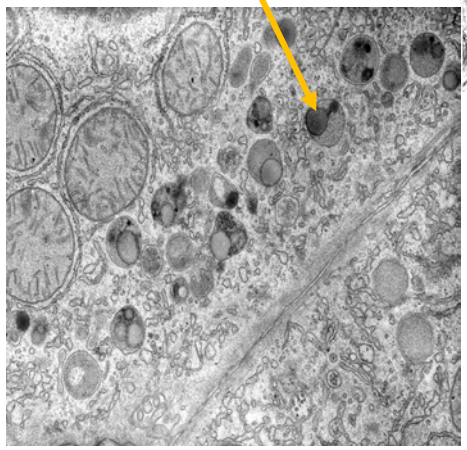


Primary lysosomes
(hydrolytic vesicles)
-pH 7
-NO DIGESTION



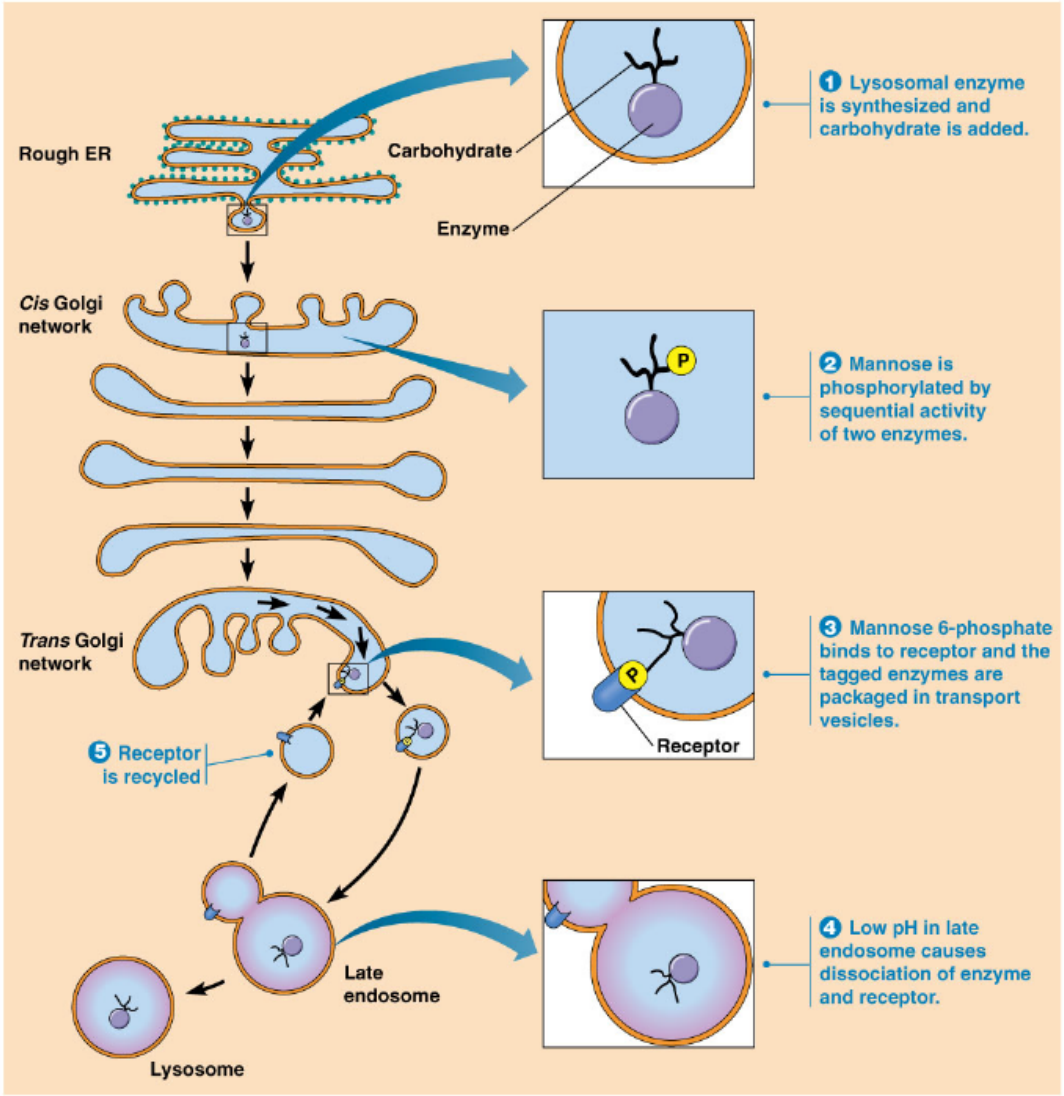
Secondary lysosomes
(Endolysosomes)
-pH 5
-DIGESTION





Morphology of lysosomes
Is very variable

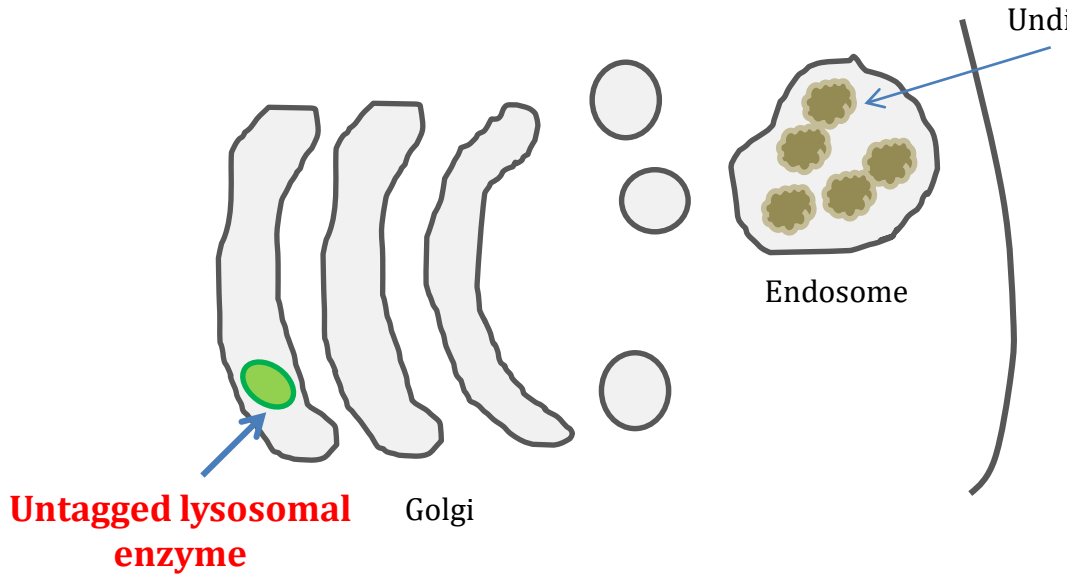
Tagging and trafficking of lysosomal enzymes



© 2012 Pearson Education, Inc.

- Synthesis/ N-glycosylation (RER)
- Phosphorylation in mannose (**M6P**)
- Recognition by a receptor
- Travel in vesicles and fusion with Late endosome

Defects in tagging/sorting



I-cell disease *inclusion-cell disease*
(mucocolipidosis-II)

-Autosomal recessive

-Lack of phosphotransferase (no M6P)

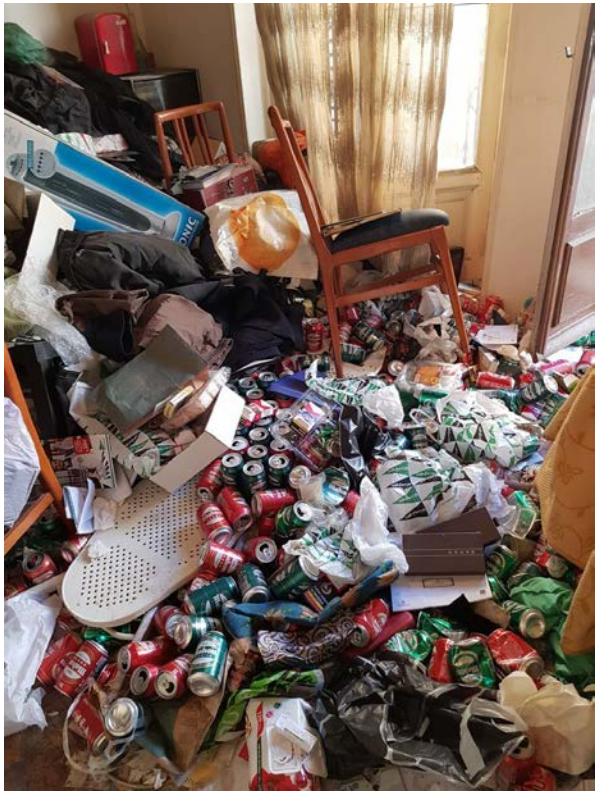
-Lysosomes devoid of enzymes

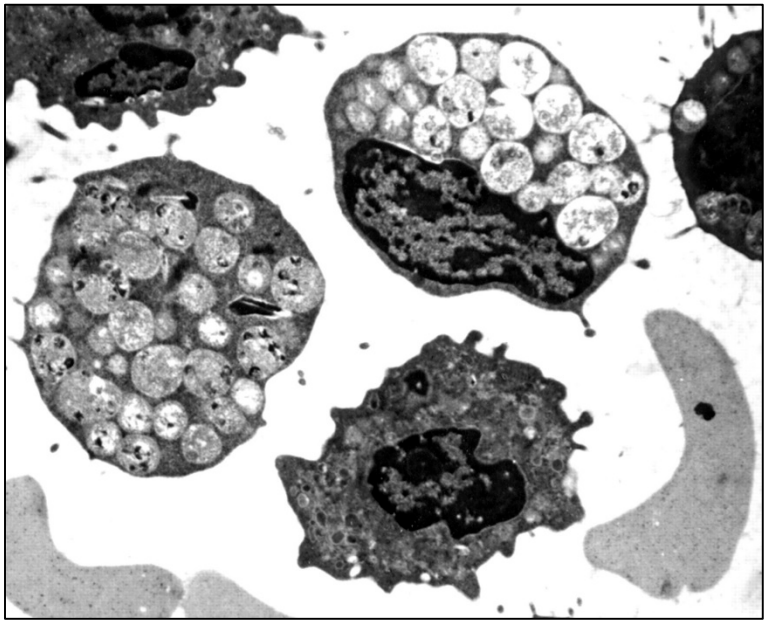
-Accumulation of undigested substrates

-Abnormal development. Dwarfism



A storage problem: accumulation of garbage…





Peripheral blood lymphocytes in I-cell disease

Lysosomal storage disorders (LSD)

-Storage of undigested substrates

-Cell stress and malfunctioning

} Developmental and metabolic problems

General features of LSD

- More than **50** different LSD
- Worldwide incidence: 1:5000 – 1:8000 newborns **(rare diseases)**

-Inheritance:

- Autosomal recessive.
- Some: X-linked (example: **Fabry**).

High incidence in endogamic populations



Ashkenazi jews (The Jewish burial society, Prague, 1772)

DOR YESHORIM
מונע דמעות
PREVENTING TEARS

Premarital counseling to prevent LSD in Jewish communities

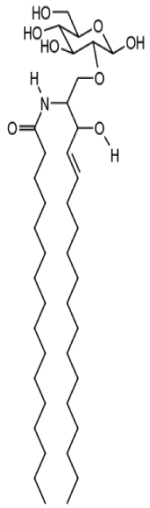
↓

Founded by Rabbi Josef Ekstein, 1983, in Brooklyn

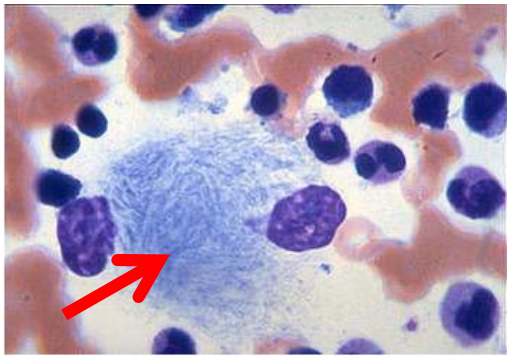
https://doryeshorim.org/

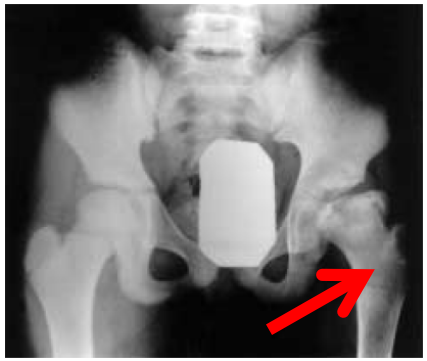
Dor Yeshorim was built on the premise that fatal and debilitating recessive genetic disorders, prevalent in Jewish circles, have absolutely no reason to be perpetuated

Gaucher



- Glucocerebrosidase deficiency
- Glucosylceramide accumulates, mainly in macrophages





-Skeletal and neurological defects

-Hepatosplenomegaly

TREATMENT

- Symptomatic treatment
- Bone marrow transplant
- Enzyme replacement therapy
- Gene therapy: *in trials*

Enzyme replacement therapy for Gaucher's disease

